

## CMOS VLSI Design Project

# Design and Analysis of a 2-Input CMOS NAND Gate

Using Cadence Virtuoso

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**Submitted By:**

Name: Piyal Chakraborty

Student ID: 2108028

**Course:** VLSI Technology Sessional

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Department of Electronics and Telecommunication Engineering, CUET

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# 1 Introduction

The CMOS NAND gate is one of the most important basic building blocks of digital integrated circuits. It performs the logical NAND operation, where the output becomes LOW only when all input signals are HIGH.

In this project, a two-input CMOS NAND gate was designed and analyzed using the Cadence Virtuoso design environment. The complete custom CMOS design flow included schematic design, symbol generation, physical layout design, Design Rule Check (DRC), Layout Versus Schematic (LVS) verification, RC parasitic extraction (RCX), and post-layout simulation.

The post-layout simulation was used to evaluate important performance parameters such as average power consumption and propagation delay.

The Boolean expression of a two-input NAND gate is

$$Y = \overline{AB} \quad (1)$$

where  $A$  and  $B$  represent the input signals and  $Y$  represents the output.

# 2 Objectives

The main objectives of this project are:

- To design a two-input CMOS NAND gate using Cadence Virtuoso.
- To create and verify the transistor-level schematic of the NAND gate.
- To generate a reusable symbol for the designed NAND gate.
- To design the physical layout of the CMOS NAND gate.
- To verify the physical layout using Design Rule Check (DRC).
- To verify the electrical equivalence between the schematic and layout using Layout Versus Schematic (LVS).
- To perform parasitic resistance and capacitance extraction using RCX.
- To perform post-layout transient simulation using the extracted circuit.
- To calculate the average power consumption of the NAND gate.
- To calculate the propagation delay of the NAND gate.

### 3 Theory of CMOS NAND Gate

A two-input CMOS NAND gate consists of two PMOS transistors forming the pull-up network and two NMOS transistors forming the pull-down network.

The two PMOS transistors are connected in parallel between  $V_{DD}$  and the output node. The two NMOS transistors are connected in series between the output node and ground.

The input signals  $A$  and  $B$  control the gates of the corresponding PMOS and NMOS transistors.

The Boolean expression of the NAND gate is

$$Y = \overline{AB}. \quad (2)$$

The truth table of the two-input CMOS NAND gate is shown in Table 1.

Table 1: Truth Table of a Two-Input CMOS NAND Gate

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

The output becomes LOW only when both input signals are HIGH.

When  $A = 1$  and  $B = 1$ , both NMOS transistors turn ON and create a conducting path from the output to ground. At the same time, both PMOS transistors turn OFF. Therefore, the output becomes LOW.

When either  $A$  or  $B$  is LOW, at least one PMOS transistor turns ON, providing a path between  $V_{DD}$  and the output. At least one NMOS transistor is also OFF, breaking the pull-down path. Therefore, the output becomes HIGH.

Hence, the implemented circuit performs the required NAND operation:

$$Y = \overline{AB}. \quad (3)$$

### 4 Transistor Sizing

The CMOS NAND gate was designed using two PMOS transistors and two NMOS transistors.

From the designed schematic, the approximate transistor dimensions used in the circuit are:

Table 2: Transistor Dimensions Used in the CMOS NAND Gate

| Transistor | Number | Width  | Length |
|------------|--------|--------|--------|
| PMOS       | 2      | 480 nm | 100 nm |
| NMOS       | 2      | 240 nm | 100 nm |

Therefore, the PMOS-to-NMOS width ratio used in the design is

$$\frac{W_p}{W_n} = \frac{480}{240} = 2. \quad (4)$$

The PMOS transistors were designed with approximately twice the width of the NMOS transistors. This provides stronger pull-up capability and helps compensate for the lower carrier mobility of holes compared with electrons.

## 5 Circuit Design

### 5.1 Schematic Design

The two-input CMOS NAND gate schematic was created using the Cadence Virtuoso Schematic Editor.

The circuit consists of two PMOS transistors connected in parallel between  $V_{DD}$  and the output node, while two NMOS transistors are connected in series between the output node and ground.

The input terminals are represented by  $A$  and  $B$ , while the final output terminal is represented by  $Y$ .

The schematic of the designed circuit is shown in Figure 1.

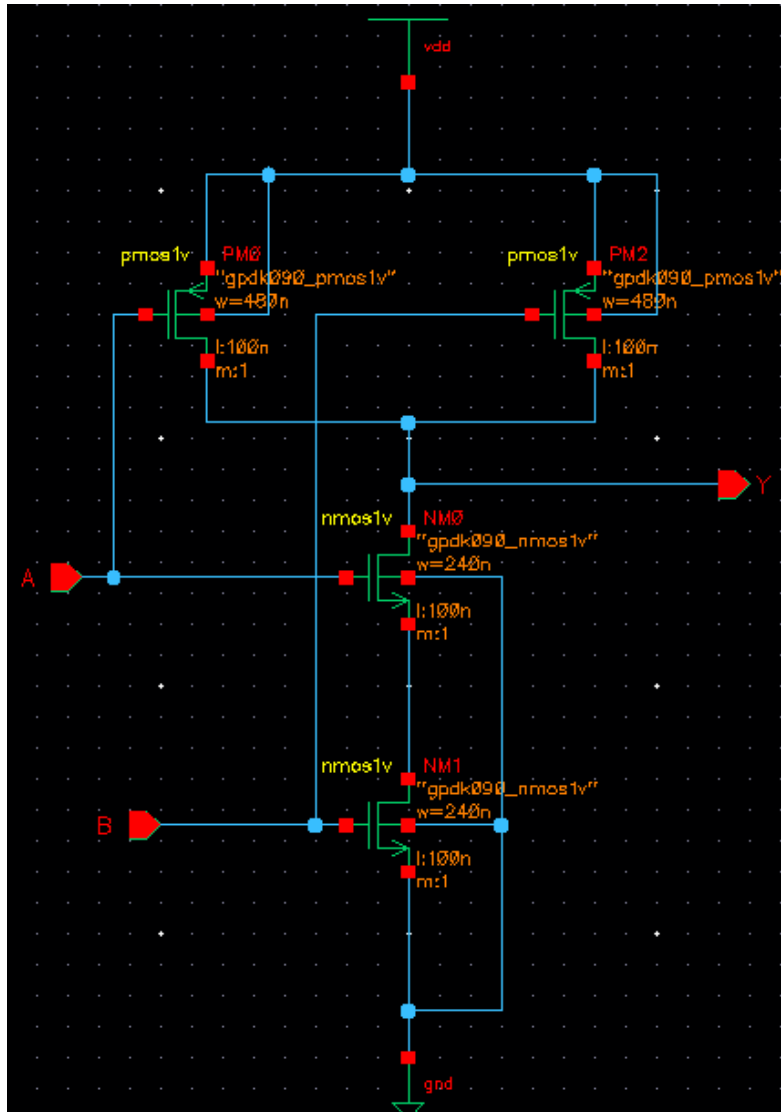


Figure 1: Schematic of the two-input CMOS NAND gate in Cadence Virtuoso.

The logical operation of the circuit can be represented as

$$A, B \rightarrow \overline{AB}. \quad (5)$$

## 5.2 NAND Gate Symbol

After completing the schematic, a symbol was generated for the two-input CMOS NAND gate.

The symbol contains two input terminals labeled  $A$  and  $B$  and one output terminal labeled  $Y$ .

The generated NAND gate symbol is shown in Figure 2.

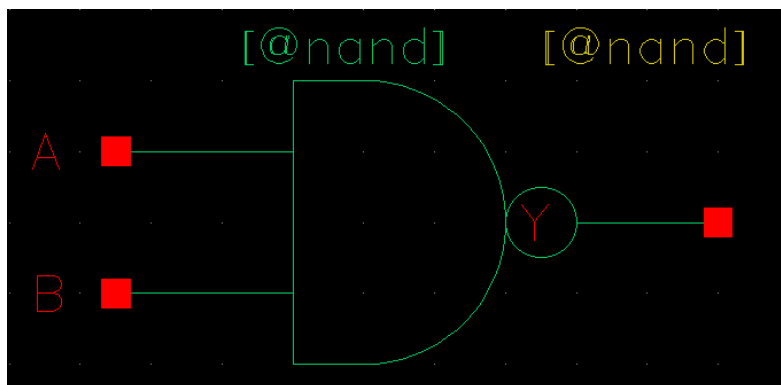


Figure 2: Generated symbol of the two-input CMOS NAND gate.

The symbol allows the designed NAND gate to be used conveniently as a reusable hierarchical block in larger digital circuits.

## 6 Physical Layout Design

After completing the schematic and symbol generation, the physical layout of the CMOS NAND gate was developed using Cadence Virtuoso Layout Editor.

The PMOS transistors were placed in the upper portion of the layout, while the NMOS transistors were placed in the lower portion.

The physical layout contains the required layers for transistor formation and electrical interconnection, including:

- active or diffusion regions,
- polysilicon,
- N-well,
- metal layers,
- contacts,
- vias,
- power connection, and
- ground connection.

The PMOS devices form the parallel pull-up network, while the NMOS devices form the series pull-down network.

The input signals are routed to the appropriate transistor gates and the output node is connected to the common drain connection of the pull-up and pull-down networks.

The completed physical layout is shown in Figure 3.

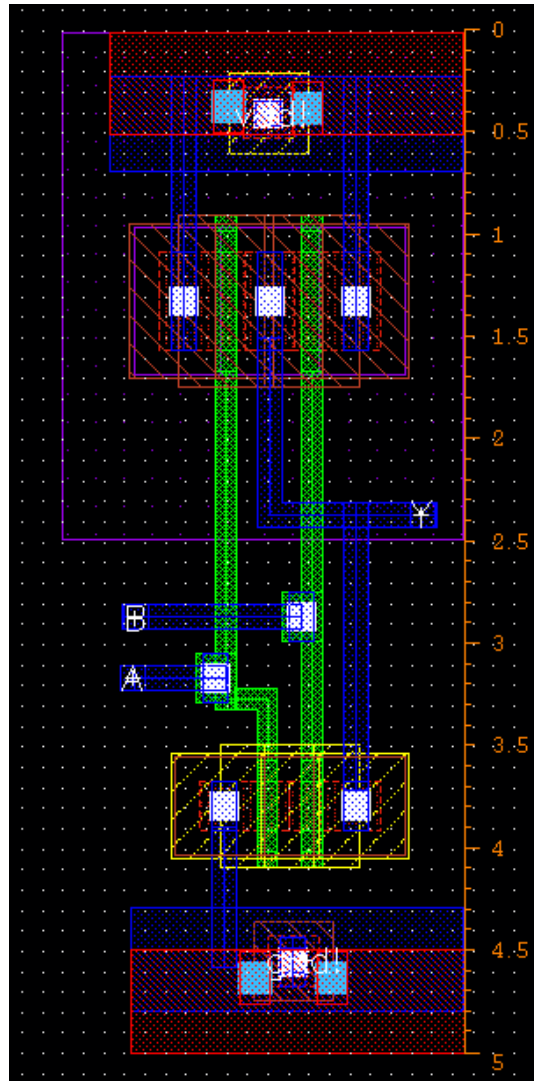


Figure 3: Physical layout of the two-input CMOS NAND gate.

The layout was designed while maintaining the required minimum width, spacing, enclosure, and connectivity rules of the selected CMOS technology.

## 7 Design Rule Check (DRC)

Design Rule Check (DRC) is performed to verify whether the physical layout satisfies the fabrication rules of the selected CMOS technology.

DRC checks various geometrical constraints such as:

- minimum metal width,
- minimum metal spacing,
- minimum polysilicon width,
- polysilicon spacing,

- diffusion spacing,
- contact dimensions,
- contact enclosure,
- via enclosure, and
- well spacing.

The completed NAND gate layout was checked using the Cadence DRC verification flow.

**DRC Result: PASSED**

The physical layout satisfies the required design rules of the selected CMOS technology.

Thus, the physical layout is valid with respect to the required fabrication design rules.

## 8 Layout Versus Schematic (LVS)

After successful DRC verification, Layout Versus Schematic (LVS) verification was performed.

LVS compares the electrical netlist extracted from the physical layout with the netlist generated from the original schematic.

It verifies important circuit properties such as:

- number of MOS transistors,
- PMOS and NMOS device types,
- gate connectivity,
- source and drain connectivity,
- input connections,
- output connection,
- power connection, and
- ground connection.

The final NAND gate layout was checked against the original schematic.

**LVS Result: PASSED**

The physical layout is electrically equivalent to the original schematic.

The successful LVS result confirms that the intended electrical connectivity was maintained during the physical layout design.

## 9 RCX and Parasitic Extraction

After DRC and LVS verification, RC parasitic extraction was performed using the RCX extraction flow.

Physical interconnections in an integrated circuit introduce additional parasitic resistance and capacitance.

These parasitic elements can affect important performance parameters such as propagation delay and power consumption.

RCX extracts these parasitic resistance and capacitance components from the physical layout.

The extracted view of the designed NAND gate is shown in Figure 4.

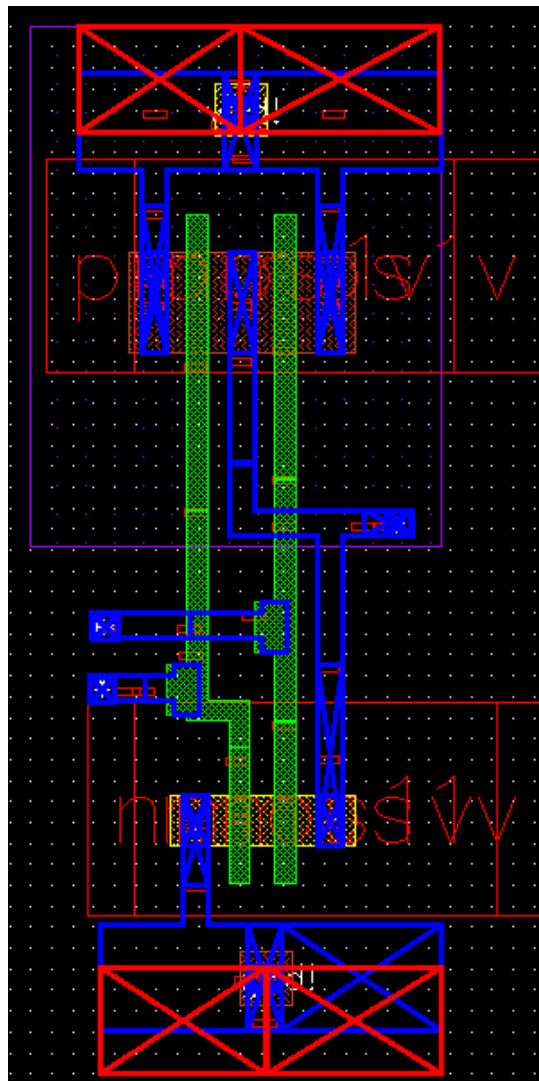


Figure 4: RCX/extracted view of the two-input CMOS NAND gate.

### RCX: Successfully Completed

The extracted NAND gate circuit was used for post-layout simulation.

## 10 Post-Layout Simulation

After parasitic extraction, post-layout transient simulation was performed using the RC-extracted view of the NAND gate.

Unlike ideal schematic-level simulation, post-layout simulation includes the effects of parasitic resistance and capacitance generated by the physical layout.

Therefore, post-layout simulation provides a more realistic estimation of the circuit performance.

The main performance parameters calculated in this project were:

1. Average power consumption
2. Propagation delay

## 11 Average Power Calculation

The average power consumption of the NAND gate was determined from the post-layout transient simulation results using the Cadence Virtuoso Calculator.

The average power over a time interval  $T$  can be expressed as

$$P_{\text{avg}} = \frac{1}{T} \int_0^T p(t) dt \quad (6)$$

where  $p(t)$  represents the instantaneous power consumption.

The average value of the simulated power waveform was calculated using the Cadence Virtuoso Calculator.

The power calculation is shown in Figure 5.

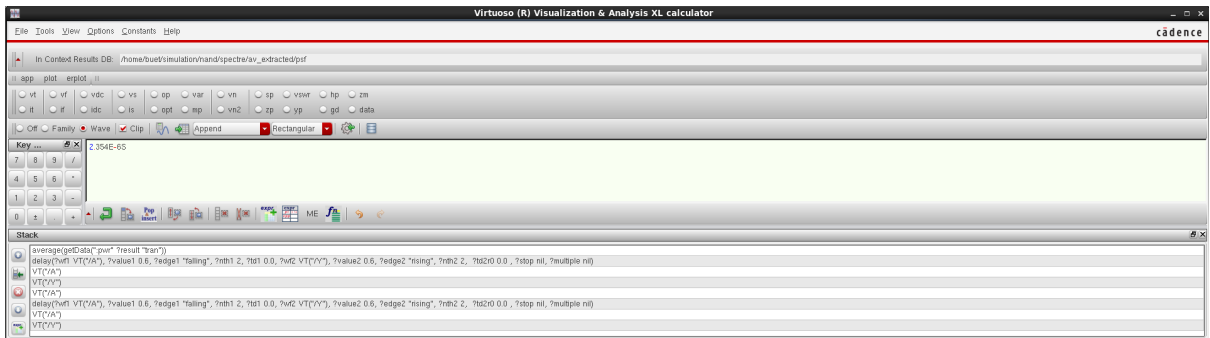


Figure 5: Average power calculation of the extracted CMOS NAND gate using the Cadence Virtuoso Calculator.

From the Cadence Calculator, the obtained average power is

$$P_{\text{avg}} = 2.354 \times 10^{-6} \text{ W.} \quad (7)$$

Since

$$1 \mu\text{W} = 10^{-6} \text{ W}, \quad (8)$$

the average power consumption can be written as

$$P_{\text{avg}} = 2.354 \mu\text{W} \quad (9)$$

Therefore, the post-layout average power consumption of the designed two-input CMOS NAND gate is approximately

$$2.354 \mu\text{W}. \quad (10)$$

## 12 Propagation Delay Calculation

Propagation delay represents the amount of time required for an input transition to produce the corresponding transition at the output of a digital circuit.

For the NAND gate, an input transition that causes the output to change is used to determine the propagation delay.

The propagation delay can generally be expressed as

$$t_p = t_{\text{out}} - t_{\text{in}}. \quad (11)$$

In this simulation, a switching reference level of approximately 0.6 V was used.

The Cadence Virtuoso Calculator was used to determine the delay between the selected input transition and the corresponding output transition.

The delay calculation is shown in Figure 6.

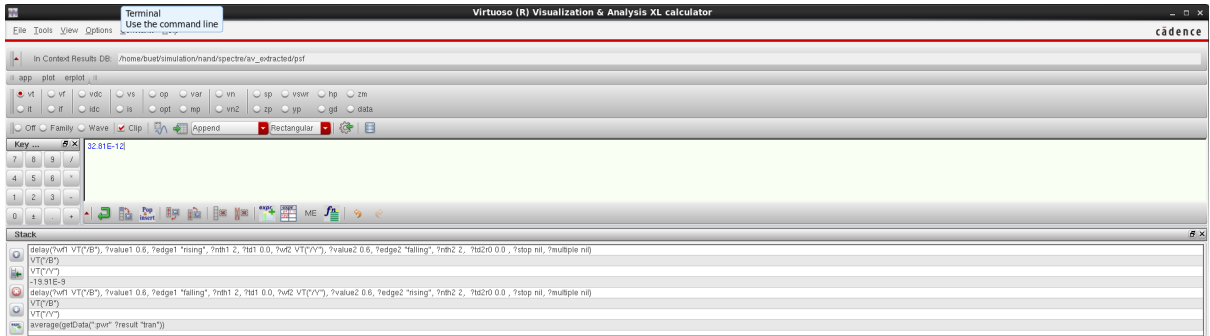


Figure 6: Propagation-delay calculation of the extracted CMOS NAND gate using the Cadence Virtuoso Calculator.

The measured propagation delay obtained from the Cadence Calculator is

$$t_p = 32.81 \times 10^{-12} \text{ s}. \quad (12)$$

Since

$$1 \text{ ps} = 10^{-12} \text{ s}, \quad (13)$$

the propagation delay becomes

$$t_p = 32.81 \text{ ps} \quad (14)$$

Therefore, the measured post-layout propagation delay for the selected input-output transition of the CMOS NAND gate is approximately

$$32.81 \text{ ps}. \quad (15)$$

## 13 Verification and Performance Summary

The overall design, verification, and performance results of the two-input CMOS NAND gate are summarized in Table 3.

Table 3: Summary of the CMOS NAND Gate Design Results

| Parameter                         | Result                                      |
|-----------------------------------|---------------------------------------------|
| <b>Logic Function</b>             | $Y = \overline{AB}$                         |
| <b>CMOS Implementation</b>        | Two PMOS in parallel and two NMOS in series |
| <b>PMOS Width</b>                 | 480 nm                                      |
| <b>NMOS Width</b>                 | 240 nm                                      |
| <b>Channel Length</b>             | 100 nm                                      |
| <b>Schematic Design</b>           | Completed                                   |
| <b>Symbol Generation</b>          | Completed                                   |
| <b>Physical Layout</b>            | Completed                                   |
| <b>DRC</b>                        | Passed                                      |
| <b>LVS</b>                        | Passed                                      |
| <b>RCX / Parasitic Extraction</b> | Completed successfully                      |
| <b>Post-layout Simulation</b>     | Completed                                   |
| <b>Average Power</b>              | $2.354 \mu\text{W}$                         |
| <b>Propagation Delay</b>          | 32.81 ps                                    |

## 14 Discussion

The complete custom CMOS design flow for a two-input NAND gate was carried out using Cadence Virtuoso.

The transistor-level circuit consists of two PMOS transistors forming the pull-up network and two NMOS transistors forming the pull-down network.

The PMOS transistors are connected in parallel, allowing the output to be pulled HIGH when either input is LOW. The NMOS transistors are connected in series, so that a complete path to ground is created only when both inputs are HIGH.

The transistor dimensions used in the design were approximately 480 nm for each PMOS transistor and 240 nm for each NMOS transistor, with a channel length of 100 nm.

After completing the schematic, a reusable NAND gate symbol was generated. The physical layout was then developed using the required active, polysilicon, well, metal, contact, via, power, and ground layers.

The layout was checked using Design Rule Check and Layout Versus Schematic verification.

RC parasitic extraction was then performed to obtain the additional resistance and capacitance associated with the physical layout. The extracted circuit was subsequently used for post-layout transient simulation.

The average post-layout power consumption obtained from the Cadence Virtuoso Calculator was approximately

$$P_{\text{avg}} = 2.354 \mu\text{W}. \quad (16)$$

The measured propagation delay for the selected input-output transition was approximately

$$t_p = 32.81 \text{ ps}. \quad (17)$$

These results include the effects of layout-dependent parasitic elements and therefore provide a more realistic indication of the circuit performance than an ideal schematic-level simulation.

## 15 Conclusion

A two-input CMOS NAND gate was successfully designed, implemented, and analyzed using Cadence Virtuoso.

The required Boolean function was

$$Y = \overline{AB}. \quad (18)$$

The NAND gate was implemented using two PMOS transistors in parallel and two NMOS transistors in series.

The project covered the major stages of the custom CMOS VLSI design flow, including schematic design, symbol generation, physical layout design, DRC verification, LVS verification, RC parasitic extraction, and post-layout simulation.

The physical layout was checked against the required design rules and its electrical connectivity was verified against the original schematic.

RCX was completed successfully, allowing the effects of parasitic resistance and capacitance to be included in the final simulation.

The obtained post-layout average power consumption was

$$\boxed{P_{\text{avg}} = 2.354 \mu\text{W}} \quad (19)$$

and the measured propagation delay was

$$\boxed{t_p = 32.81 \text{ ps.}} \quad (20)$$

Therefore, the project demonstrates the complete schematic-to-layout design, verification, extraction, and performance analysis of a two-input CMOS NAND gate using Cadence Virtuoso.

## 16 Software Used

The following software tools were used to complete the project:

- Cadence Virtuoso Schematic Editor
- Cadence Virtuoso Layout Editor
- Cadence Spectre Simulator
- Design Rule Check (DRC) Tool
- Layout Versus Schematic (LVS) Tool
- RCX / Parasitic Extraction Tool
- Cadence Virtuoso Visualization and Analysis Calculator